

DHRUV RAJAK

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Education

Indian Institute of Information Technology Lucknow

Bachelor of Technology in Computer Science Engineering

Nov 2020 – May 2024

Lucknow, Uttar Pradesh

Skills

Languages: JavaScript, TypeScript, HTML, CSS, Bash

Front End: ReactJS, React Native, Expo, NextJS, Vite, Tailwind, Zustand, TanStack Query

Back End: NodeJS, REST-API, Hono, tRPC, Mongoose, MySQL, SQLite/D1, AWS Lambda, AWS DynamoDB, Redis/MemoryDB

Auth & Infra: OAuth 2.0, Cloudflare Workers/Pages/D1, AWS, Firebase

Version Control: Git/GitHub

Developer Tools: VS Code, Postman, MongoDB Atlas, Figma, RevenueCat

Experience

BharatBrainz by Vyas Edification

Jan 2024 – current

Founding Developer on a 3-person team (ReactJs, React Native, Node.js, AWS)



- Founding developer on a 3-person team that took BharatBrainz from zero to 200+ daily active users across Play Store and App Store.
- Reduced page-load time by 80% (Flashlight metrics) on the BharatBrainz React Native app via image optimization, re-render minimization, and FlatList optimization.
- Reduced AWS costs by 28% year-over-year by fixing API-fetch and query mechanisms, S3 lifecycle rule configuration to cut retrieval times, eliminate redundant calls, and optimize S3 storage and retrieval costs.
- Engineered a review-and-finalization pipeline enabling 200+ uploaders to submit 50k+ custom questions and 10k+ QuestionSets, cutting administrative overhead by 50%.
- Built core app architecture including custom video player, route architecture, Firebase Analytics, persistent storage (Zustand+AsyncStorage), and WebViews for LaTeX rendering.
- Streamlined content-generation workflows (QuestionSet/Question/Solution generation, weak-area recommendation) using Gemini VertexAI, and deployed background cron jobs like weekly report generation and mailing for Classroomz.
- Shipped the vED website for course sales with SEO best practices and account management in a 2-week sprint.

Faxify by ConstAgility

Dec 2023 – Feb 2024

Full-Stack Freelancer

- Delivered drag-and-drop fax upload and delivery features for the Faxify web app, streamlining the end-to-end sending workflow.
- Reduced resource costs by implementing PDF compression, image manipulation, and file-to-PDF conversion, alongside dependency upgrades and bug fixes across the codebase.

Projects

Buzzwordle | *React 19, TypeScript, Cloudflare Workers, D1, Hono*

June 2026

- Implemented Google OAuth 2.0 with cookie-based sessions and anonymous-to-account session merging on login, including conflict resolution for already-completed words and cross-device duplicate-attempt guards to keep records consistent across concurrent sessions.
- Architected a serverless edge backend (Cloudflare Workers + D1 via Hono, Chanfana for OpenAPI/Zod validation) with automated cron-based cleanup of incomplete anonymous sessions, and a submission-triggered leaderboard backed by indexed queries for fast ranking retrieval.
- Built an admin platform with calendar-based word scheduling (pattern-based bulk fill, manual override) and an AI-conditioned content engine generating per-word hints and contextual success/fail messages, with individual regeneration and manual override.
- Created analytics dashboards (admin + personal) covering session funnels, win/fail rates, and a monthly word-difficulty ranking gated by minimum attempt count to avoid low-sample skew.

CobaltDocs - privacy focused document manager | *React Native, Zustand, SQLite, Encryption*

October 2025

- Developed a secure document management mobile app with document listing, card-based UI, and categorized semantic-search, improving document access efficiency.
- Implemented AES-based file encryption/decryption, master key generation, and secure local storage for 100% on-device data protection.
- Constructed an end-to-end document capture flow (camera → edit → encrypt → save) covering 4-step upload pipeline (with photo editing & filter pipeline).
- Designed SQLite-based local database with schema initialization for persistent and secure storage.